

SEAT No. _____



Total No. of printed pages : 03

[81382]

SARDAR PATEL UNIVERSITY

M. Sc. Information Technology, Semester-1 Examinations

PS01CINT51/ PS01CINT31 : Object Oriented Programming Using Java

Date: 11-02-2022, Friday Time: 12:00 pm to 2:00 pm

Total Marks: 70

Q.1 A Multiple Choice Questions.

[08]

- 1 What is the extension of a java code?
A. .java B. .class
C. .obj D. none of these
- 2 Which of the following is not possible to define in a typical java application directly?
A. primitive data such as integer B. pointer
C. user defined data such as student D. none of these
- 3 _____ keyword is used to inherit a class.
A. implements B. extended
C. extends D. inherits
- 4 Java supports _____.
A. function overloading B. local variable
C. multi level inheritance D. all of above
- 5 _____ occurs if a program tries to access an array index that is negative, greater than, or equal to the length of the array.
A. ArrayIndexOutOfBoundsException B. ArithmeticException
C. IOException D. none of these
- 6 _____ is a passive control.
A. JTextField B. JLabel
C. JButton D. JComboBox
- 7 Which of these keywords is used to define interfaces in Java?
A. extends B. extended
C. interface D. none of these
- 8 JDBC application must import _____ package.
A. string B. data
C. math D. none of these

[13]

CP.T.O.3

Q.1 B Do as Directed.

[16]

- 1 List only any two features of java.
- 2 Give full form of JVM.
- 3 State true or false: Java bytecode is interpreted by the Java Virtual Machine.
- 4 State true or false: The super keyword in Java is a reference variable which is used to refer immediate parent class object.
- 5 What is an object class in java?
- 6 List three basic access specifiers in java.
- 7 What will be output of the following java block?

```
class Increment {  
    public static void main(String args[])  
    {    int g = 3; System.out.print(++g * 8);    } }
```
- 8 Which java class provides system independent server side implementation?
- 9 What will be output of the following java block?

```
class Temp  
{ public static void main(String args[])  
  { StringBuffer s1 = new StringBuffer("temp");  
    StringBuffer s2 = s1.reverse(); System.out.println(s2);  
  } }.
```
- 10 Which method of the class String is used to obtain a length of String Object?
- 11 Give full form of AWT.
- 12 Give full form of JFC.
- 13 Give full form of JDBC.
- 14 State true or false: Servlets execute within the address space of web server, platform independent and uses the functionality of java class libraries.
- 15 _____ is a GUI widget toolkit for Java. It is part of Oracle's JFC – an API for providing a graphical user interface for Java programs.
- 16 State true or false: The `itemStateChanged()` method is invoked automatically whenever you click or unclick on the registered checkbox component.

Q.2 Short Questions (Attempt any Seven) [14]

- 1 Give the anatomy of a simple java program.
- 2 Give an example of if statement in java.
- 3 Define a single dimension array of 5 integers in java.
- 4 Define a constant Pi=3.14 in java.
- 5 Define constructor of an employ class. The employ class has two string type variables employid and employname.
- 6 Define polymorphism. Also give a brief example of polymorphism in java.
- 7 Explain any two string methods.
- 8 List only any two methods of Reader class.
- 9 List only any two member functions from the java util package.

Q.3 Explain how java offers platform independence. [08]

OR

Q.3 Explain (i) for loop and (ii) while loop in java by taking suitable examples. [08]

Q.4 What is an exception? Explain the syntax of try and catch block with a suitable example. [08]

OR

Q.4 Write a note on Multithreading in java. [08]

Q.5 What is a Layout? List and explain different layouts available in a swing. [08]

OR

Q.5 Write a short note on the following components with properties and methods. [08]

1. JTextField 2. JButton

Q.6 How a java program can interact with database? Explain steps to get a database connection by taking a simple java program. [08]

OR

Q.6 Explain common JDBC components such as DriverManager, Driver, Connection, Statement, etc. in brief. [08]

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